X – IIT FOUNDATION | OLYMPIAD · MATHS VOL-2

Parabola

A step-by-step guide: what it is, where the equation comes from, and how to use it

CONCEPT 1

What is a conic?

IN SIMPLE WORDS

Pick a point S (focus) and a line l (directrix). For any point P, compare its distance to S with its distance to l. If the ratio is always the same number, P traces a conic.

SP / PM = e (constant)

- SP = distance from focus; PM = perpendicular distance to directrix.
- e is the **eccentricity**.
- **e = 1** → parabola
- **e < 1** → ellipse
- **e > 1** → hyperbola

CONCEPT 1

Equation of a conic: how it is built

IN SIMPLE WORDS

Write “$SP^2 = e^2 \cdot PM^2$” in coordinates, expand, and collect terms. The result always has x^2, xy, y^2, x, y and a constant.

$$PM = (px_1+qy_1+r)/\sqrt{(p^2+q^2)}$$

$$(x_1-a)^2 + (y_1-b)^2 = e^2(px_1+qy_1+r)^2/(p^2+q^2)$$

- Focus S(a,b); directrix $px + qy + r = 0$; $P(x_1,y_1)$ on the conic.
- Simplified: $(1 - p^2e^2/(p^2+q^2))x^2 + (1 - q^2e^2/(p^2+q^2))y^2 - 2pqe^2/(p^2+q^2)\cdot xy - 2(a + pre^2/(p^2+q^2))x - 2(b + qre^2/(p^2+q^2))y + (a^2+b^2-r^2e^2/(p^2+q^2)) = 0$ …(1)
- This is $\mathbf{Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0}$: a conic is a 2nd-degree equation (converse proved later).

CONCEPT 1

Parabola: definition

IN SIMPLE WORDS

A parabola is the special conic with e = 1: every point is exactly as far from the focus as from the directrix.

SP = PM

- Put e = 1 in (1): $(qx - py)^2/(p^2+q^2) - 2(a + pr/(p^2+q^2))x - 2(b + qr/(p^2+q^2))y + (a^2+b^2-r^2/(p^2+q^2)) = 0$.
- Its second-degree terms form a **perfect square**: a quick test for a parabola.

CONCEPT 1

Deriving $y^2 = 4ax$

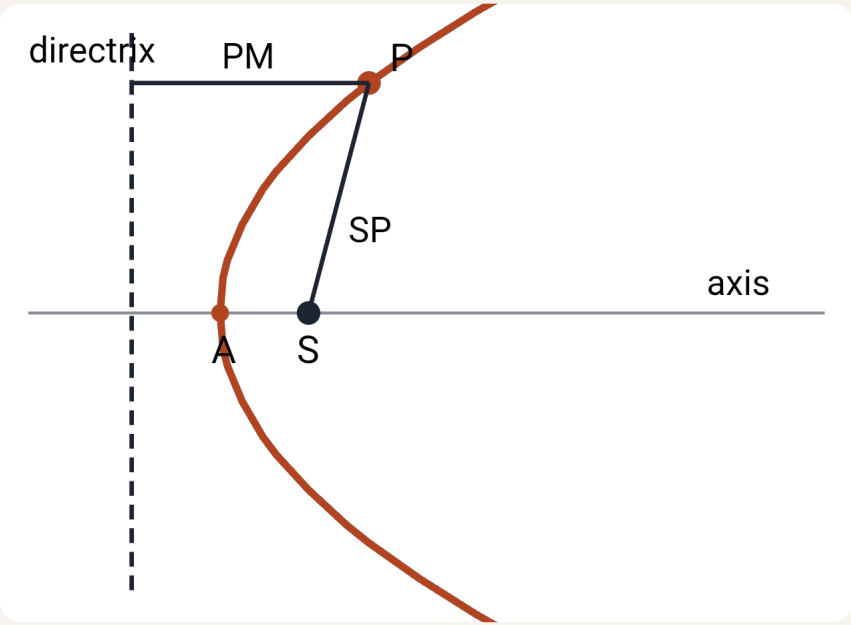
IN SIMPLE WORDS

Choose axes smartly: put the vertex at the origin and the axis along the x-axis. The equation then becomes very simple.

$$(x-a)^2 + y^2 = (x+a)^2$$

$$\Rightarrow y^2 = 4ax$$

- S = focus, Z = foot of S on directrix l. A = midpoint of SZ, SA = AZ = a, so SA/AZ = 1 = e: A is on the parabola.
- Take AS as x-axis, AY ∥ l as y-axis: A(0,0), S(a,0), Z(−a,0).
- $SP^2 = MP^2$ gives the result.
- **Vertex** A(0,0); **focus** S(a,0); **directrix** x + a = 0.



CONCEPT 1

Shape of $y^2 = 4ax$ ($a > 0$)

IN SIMPLE WORDS

Read the equation to “see” the curve: signs and symmetry tell you where it lives and how it behaves.

$$y^2 = 4ax$$

- (x,y) on C $\Leftrightarrow$ $(x,-y)$ on C: symmetric about the x-axis (the **axis**).
- $y = 0 \Rightarrow x = 0$: meets the axis only at (0,0).
- $x > 0 \Rightarrow$ y has two real values: lies in quadrants I and IV.
- $x = 0 \Rightarrow y = 0$,0: the y-axis touches it at the origin.
- $x < 0 \Rightarrow$ no real y: nothing in quadrants II, III.
- $x \to +\infty \Rightarrow y \to \pm\infty$: not bounded on the right.
- $P(x_1,y_1)$ is external/internal as $y_1^2 >$ or $< 4ax_1$.

CONCEPT 1

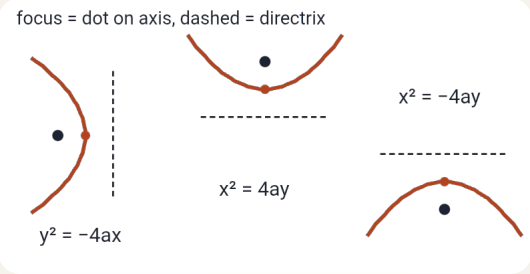
Other forms: same shape, different position

IN SIMPLE WORDS

Rotate or shift the standard parabola and only the focus, vertex and directrix change.

$$(y-\beta)^2 = 4a(x-\alpha)$$

$$(x-\alpha)^2 = 4a(y-\beta)$$



- $y^2 = -4ax$: focus $(-a,0)$, directrix $x - a = 0$.
- $x^2 = 4ay$: focus $(0,a)$, directrix $y + a = 0$.
- $x^2 = -4ay$: focus $(0,-a)$, directrix $y - a = 0$.
- $(y-\beta)^2 = 4a(x-\alpha)$: axis ∥ x-axis, vertex (α,β), focus $(\alpha+a,\beta)$, directrix $x = \alpha - a$; reducible to $x = ly^2 + my + n$.
- $(x-\alpha)^2 = 4a(y-\beta)$: axis ∥ y-axis, vertex (α,β), focus $(\alpha,\beta+a)$, directrix $y = \beta - a$.
- $(x-a)^2 + (y-b)^2 = (lx+my+n)^2/(l^2+m^2)$: focus (a,b), directrix $lx+my+n = 0$, axis $m(x-a) - l(y-b) = 0$.

CONCEPT 1

Focal distance, focal chord, double ordinate

IN SIMPLE WORDS

Three names to learn: distance to the focus, chords through the focus, and chords perpendicular to the axis.

Focal distance of $P(x_1,y_1) = x_1 + a$

- **Focal distance:** SP. Since SP = PM = NZ = NA + AZ = x_1 + a. Equals abscissa of P + ¼(latusrectum).
- **Focal chord:** chord drawn through the focus.
- **Double ordinate** of P: chord through P perpendicular to the axis.

CONCEPT 1

Latusrectum

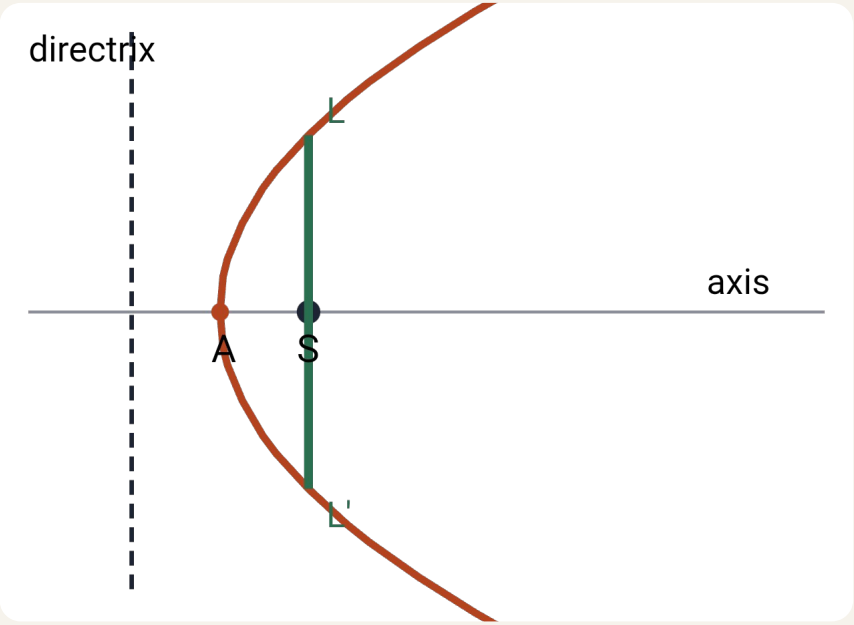
IN SIMPLE WORDS

The latusrectum is the double ordinate that passes through the focus. Its length is found by putting x = a in the equation.

$L = (a, l)$: $l^2 = 4a^2 \Rightarrow l = \pm 2a$

$LL' = 2l = 4a$

- LL′ = chord through S, perpendicular to the axis.
- Length = 4a = 4 × (distance of focus from vertex).
- Its line is x − a = 0.
- Hence the double ordinate through the focus is the latusrectum.



CONCEPT 2

Parametric form and chord

IN SIMPLE WORDS

Instead of x and y, describe each point with one number t. Then chords and tangents become easy algebra.

$(at^2, 2at)$

Chord t_1,t_2: $y(t_1+t_2) - 2x - 2at_1t_2 = 0$

- $(at^2, 2at)$ satisfies $y^2 = 4ax$ for every t.
- Proof: $(y-2at_1)/(2at_1-2at_2) = (x-at_1^2)/(at_1^2-at_2^2)$; cancel $t_1-t_2 \Rightarrow (y-2at_1)/2 = (x-at_1^2)/(t_1+t_2)$.
- **Focal chord:** passes through $(a,0) \Rightarrow 0 - 2a - 2at_1t_2 = 0 \Rightarrow \mathbf{t_1t_2 = -1}$ (remember).

CONCEPT 2

Tangent at t

IN SIMPLE WORDS

A tangent is a line with the same slope as the curve at that point. Differentiate to get the slope.

Tangent at t: $yt - x - at^2 = 0$

Tangents at t_1, t_2 meet at $(at_1t_2, a(t_1+t_2))$

- $dy/dx = 2a/y$, so slope at $(at^2, 2at) = 2a/2at = 1/t$.
- Line: $y - 2at = (1/t)(x - at^2) \Rightarrow yt - x - at^2 = 0$.
- Solve $yt_1 - x - at_1^2 = 0$ and $yt_2 - x - at_2^2 = 0$ for the intersection.

CONCEPT 2

Normal at t

IN SIMPLE WORDS

The normal is perpendicular to the tangent, so its slope is $-t$ (negative reciprocal of $1/t$).

Normal at t: $y + xt = 2at + at^3$

$t_2 = t_1 - 2/t_1$

- Slope of normal $= -t \Rightarrow y - 2at = -t(x - at^2) \Rightarrow y + xt = 2at + at^3$.
- **Where it meets the curve again:** normal at t_1 passes through $(at_2^2, 2at_2)$: $2(t_2 - t_1) = t_1(t_1^2 - t_2^2) \Rightarrow -2/t_1 = t_1 + t_2 \Rightarrow t_2 = -t_1 - 2/t_1$.

CONCEPT 2

How many normals from a point?

IN SIMPLE WORDS

Substitute (x_1,y_1) into the normal equation: you get a cubic in t. A cubic has at most three real roots, so at most three normals.

$$at^3 + t(2a - x_1) - y_1 = 0$$

$$t_1+t_2+t_3 = 0$$

$$\Sigma t_1t_2 = (2a-x_1)/a;\ t_1t_2t_3 = y_1/a$$

- $y_1 + x_1t = 2at + at^3 \Rightarrow at^3 + t(2a-x_1) - y_1 = 0$.
- (a) P internal: three normals.
- (b) P on the parabola: two normals.
- (c) P external: one normal.

CONCEPT 3

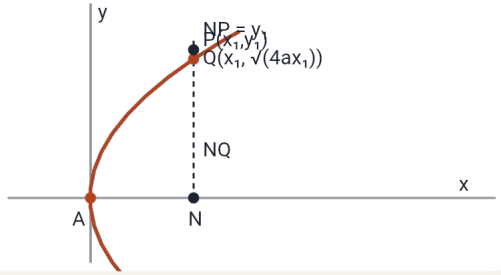
Notation and position of a point

IN SIMPLE WORDS

Plug the point into $y^2 - 4ax$ and look at the sign: it tells inside, on, or outside.

$$S = y^2 - 4ax;\ S_1 = yy_1 - 2a(x+x_1)$$

$$S_{12} = y_1y_2 - 2a(x_1+x_2);\ S_{11} = y_1^2 - 4ax_1$$



- $N=(x_1,0)$, $Q=(x_1,\sqrt{(4ax_1)})$; $NP = y_1$, $NQ = \sqrt{(4ax_1)}$.
- $\mathbf{S_{11} > 0}$ (NP > NQ): P outside.
- $\mathbf{S_{11} = 0}$ (NP = NQ): P on the parabola.
- $\mathbf{S_{11} < 0}$ (NP < NQ): P inside.

CONCEPT 3

Chord and tangent from S_1, S_2, S_{12}

IN SIMPLE WORDS

Two points on the parabola give a chord with a neat formula. Let one point slide into the other and the chord becomes a tangent.

Chord: $S_1 + S_2 = S_{12}$

Tangent at (x_1,y_1): $S_1 = 0$, i.e. $yy_1 - 2a(x+x_1) = 0$

- Proof: $S_{11} = S_{22} = 0$; $S_1+S_2 = S_{12}$ is first degree (a line).
- Put (x_1,y_1): $S_{11} + S_{12} = S_{12}$ ✓. Same for (x_2,y_2).
- As Q → P: $2S_1 = S_{11} = 0 \Rightarrow S_1 = 0$.
- Check by calculus: slope $2a/y_1 \Rightarrow y - y_1 = (2a/y_1)(x-x_1) \Rightarrow yy_1 - 4ax_1 = 2ax - 2ax_1 \Rightarrow S_1 = 0$.
- If $y_1 = 0$ then $x_1 = 0$: tangent at vertex is the y-axis.

CONCEPT 3

Tangent in slope form

IN SIMPLE WORDS

Often you know the slope, not the point. Then the tangent must have a fixed intercept c = a/m.

$y = mx + a/m \ (m \neq 0)$

Contact point $(a/m^2, 2a/m)$

- Tangent at (x_1,y_1): $yy_1 - 2a(x+x_1) = 0 \Rightarrow y_1/1 = 2a/m = 2ax_1/c \Rightarrow x_1 = c/m$, $y_1 = 2a/m$.
- On the parabola: $4a^2/m^2 = 4a(c/m) \Rightarrow \mathbf{c = a/m}$.
- **Line $lx + my + n = 0$ $(l \neq 0)$ touches iff $ln = am^2$**; contact $(n/l, -2am/l)$. Proof: compare with $yy_1 - 2a(x+x_1) = 0$: $l/-2a = m/y_1 = n/-2ax_1$.

CONCEPT 3

Lines that are NOT tangents

IN SIMPLE WORDS

Quick checks to eliminate special lines.

$$y = mx + c$$

- $m = 0$, $c \neq 0$: $y = c$ meets at $(c^2/4a, c)$: not a tangent. Any line parallel to the axis meets in one point.
- $m \neq 0$, $c = 0$: meets at $(0,0)$ and $(4a/m^2, 4a/m)$: not a tangent.
- $m = 0$, $c = 0$: $y = 0$ cuts at $(0,0)$: not a tangent.
- $x = c$: real points for $c > 0$; touches for $c = 0$.

CONCEPT 3

Normal with given slope m

IN SIMPLE WORDS

Slope of tangent = $2a/y_1$; normal slope $m = y_1/2a \Rightarrow y_1 = -2am$, $x_1 = am^2$.

$$y = mx - 2am - am^3$$

Foot: $(am^2, -2am)$

- Using $y_1^2 = 4ax_1$: $4a^2m^2 = 4ax_1 \Rightarrow x_1 = am^2$.
- Equation: $y + 2am = m(x - am^2)$.
- **Example:** $y^2 = 8x$, $a = 2$, $m = 2/3$: $y = (2/3)x - 4(2/3) - 2(8/27) \Rightarrow$ **$18x - 9y - 88 = 0$**.

CONCEPT 3

Tangents from an external point

IN SIMPLE WORDS

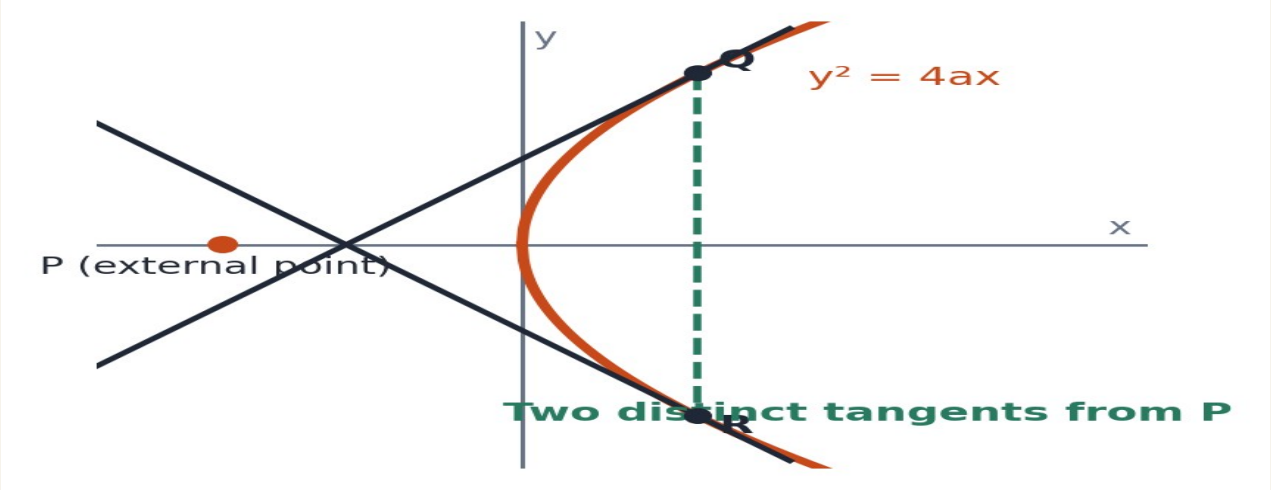
Put the point into y = mx + a/m: you get a quadratic in m. Its discriminant decides how many tangents.

$$m^2x_1 - my_1 + a = 0$$

Pair of tangents: $S_1^2 = S \cdot S_{11}$

- $y_1^2 - 4ax_1 > 0$ (outside): two real tangents.
- $= 0$: on the parabola, coincident tangents.
- < 0 (inside): no real tangent.
- Perpendicular tangents: $m_1m_2 = a/x_1 = -1 \Rightarrow x_1 + a = 0$: locus is the **directrix**.
- **Example:** from $(-2,1)$ to $y^2 = 2x$: $(y-x+2)^2 = 5(y^2-2x) \Rightarrow x^2 - 2xy - 4y^2 + 6x + 4y + 4 = 0$.

GEOMETRY: TWO TANGENTS FROM AN EXTERNAL POINT



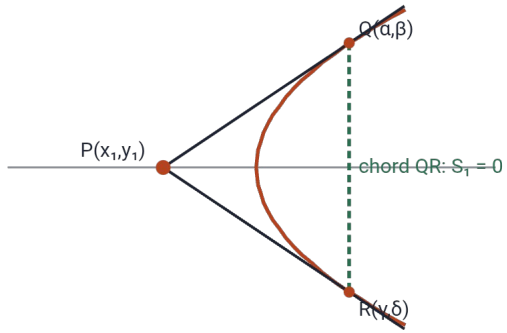
CONCEPT 3

Chord of contact

IN SIMPLE WORDS

The line joining the two points where tangents from P touch the parabola.

Chord of contact of (x_1,y_1): $S_1 = 0$



- Tangents from $P(x_1,y_1)$ touch at $Q(\alpha,\beta)$, $R(\gamma,\delta)$.
- Tangent at Q: $y\beta - 2a(x+\alpha) = 0$ passes through P: $y_1\beta - 2a(x_1+\alpha) = 0$ …(1).
- Similarly $y_1\delta - 2a(x_1+\gamma) = 0$ …(2).
- So Q and R satisfy $yy_1 - 2a(x+x_1) = 0$: that is QR.

CONCEPT 4

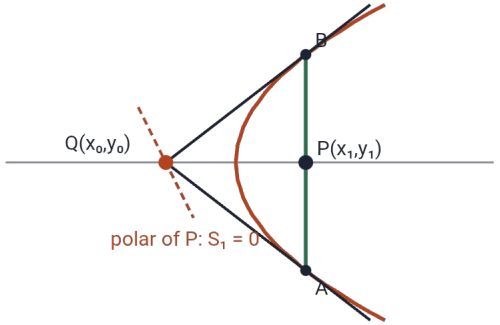
Pole and polar

IN SIMPLE WORDS

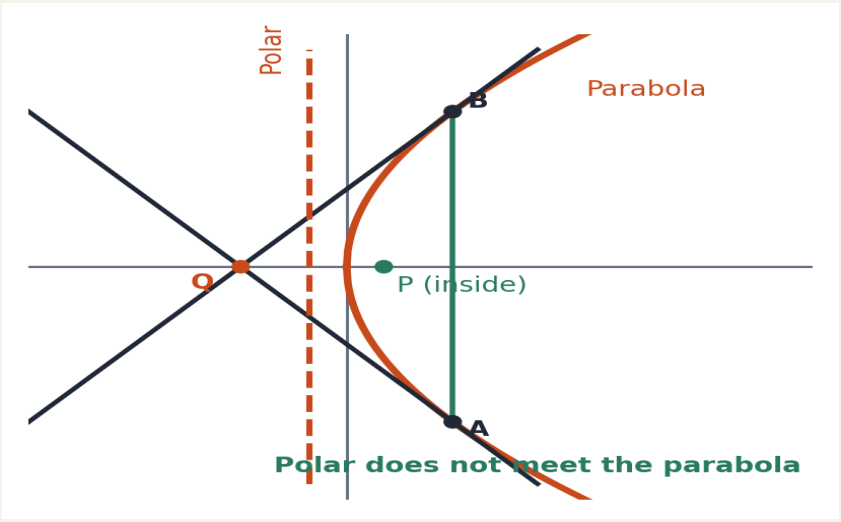
The polar of a point is the line formed by where tangents at the ends of chords through the point meet. The point is the pole.

Polar of (x_1,y_1): $S_1 = 0$

- Chord AB through P; tangents at A, B meet at $Q(x_0,y_0)$. AB is the chord of contact $S_0 = 0$ and passes through P: $y_1y_0 - 2a(x_1+x_0) = 0$. Locus of Q: $S_1 = 0$.
- P outside: polar meets the parabola in real points.
- P inside: polar does not meet it.
- P on it: polar is the tangent at P.
- Example: $y(-3) - 1(x+2) = 0 \Rightarrow x + 3y + 2 = 0$.



CASE (b): P INSIDE — POLAR DOES NOT INTERSECT



CONCEPT 4

Conjugate points, lines and pole of a line

IN SIMPLE WORDS

Two points are conjugate if each lies on the polar of the other. The same idea works for lines.

Conjugate points: $S_{12} = 0$

Pole of $lx+my+n=0$: $(n/l, -2am/l)$

Conjugate lines: $l_1n_2 + l_2n_1 = 2am_1m_2$

- Polar of $P(x_1,y_1)$ through $Q(x_2,y_2)$: $S_{21} = 0 \Rightarrow S_{12} = 0$, so polar of Q passes through P.
- Pole of a line: match $lx+my+n = 0$ with $yy_1 - 2a(x+x_1) = 0$.
- **Example:** pole of $x - y - 2 = 0$ w.r.t. $y^2 = 6x$ $(a = 3/2) = (-2, 3)$.
- Conjugate lines: pole of line 1 is $(n_1/l_1, -2am_1/l_1)$; on line 2 $\Rightarrow$ condition; it is symmetric.

CONCEPT 4

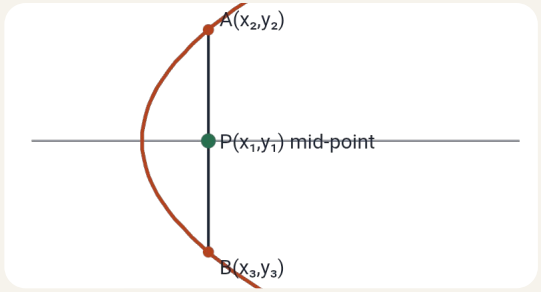
Chord with a given mid-point

IN SIMPLE WORDS

If P is the mid-point of a chord, the chord equation is $S_1 = S_{11}$.

$$S_1 = S_{11}$$

$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$



- **Example:** $y^2 = 2x$, mid-point (1,1): $y(1) - (x+1) = 1^2 - 2(1) \Rightarrow \mathbf{x - y = 0}$.

REVISION

Formula summary

IN SIMPLE WORDS
Everything in one place.

$y^2 = 4ax$; vertex (0,0); focus (a,0); directrix $x = -a$

Latusrectum 4a; focal distance $x_1 + a$

- Parametric $(at^2, 2at)$; tangent $yt - x - at^2 = 0$; normal $y + xt = 2at + at^3$.
- Tangent $S_1 = 0$; chord of contact $S_1 = 0$; polar $S_1 = 0$; pair of tangents $S_1^2 = SS_{11}$; mid-point chord $S_1 = S_{11}$.
- Tangent condition $c = a/m$; $ln = am^2$. Focal chord $t_1t_2 = -1$.